

CERTIFICATE OF RADIOLOGICAL VERIFICATION AND DATASET APPROVAL

This is to certify that the radiological dataset review, pelvimetric data verification, and pelvic floor morphometric assessment for the Ph.D. thesis:

"A COMPUTED TOMOGRAPHY EVALUATION OF PELVIC FLOOR THICKNESS AND BONY DIMENSIONS AMONG WOMEN FROM HILLY AND INLAND REGIONS OF SOUTH INDIA: A COMPARATIVE CROSS-SECTIONAL STUDY"

submitted by Mr. MUTHUKUMAR C, Research Scholar in Anatomy, USN No. 24PH14ANA01, Sri Siddhartha Academy of Higher Education (SSAHE), Tumakuru, has been thoroughly reviewed and counter-checked in the Department of Radiodiagnosis.

The candidate has compiled a structured CT pelvimetry dataset of 10 female patient studies (5 Hilly-region + 5 Inland/Plain-region) sourced from PACS repositories across institutions in Kerala, Karnataka, and Tamil Nadu. Each entry includes anonymised DICOM volumes, segmented structures, and a CT Pelvimetry & Pelvic Floor Assessment Report (PDF).

I hereby confirm and certify that:

- 1. Bony Pelvimetry Validation:** All linear and angular morphometric parameters (True/Obstetric/Diagonal Conjugate, Transverse Inlet, ISD, ITD, Subpubic Angle, Sacral Length & Curvature, Pelvic Inclination, Oblique Diameter, AP Mid-pelvis) were measured at correct anatomical planes and fall within clinically established reference ranges for the South Indian female population.
- 2. Pelvic Floor Soft-Tissue Assessment:** Bilateral thickness values for puborectalis, pubococcygeus, iliococcygeus, and coccygeus; levator hiatus AP/lateral dimensions; hiatal area; resting, Valsalva, and contraction thickness indices have been reviewed and verified for plausibility against published normative data.
- 3. Anatomical Edge Validation:** The verification of soft-tissue boundary extraction for the pelvic floor musculature (specifically the puborectalis, pubococcygeus, iliococcygeus, and coccygeus muscles) was monitored to ensure the clean exclusion of surrounding adipose tissue components.
- 4. Methodological Rigor:** The linear and angular morphometric parameters - including the Interspinous Diameter (ISD), Intertuberous Diameter (ITD), Subpubic Angle (SPA), Anteroposterior pelvic outlet diameter, and the Area of the Perineum - were measured at the correct predefined cross-sectional anatomical planes.
- 5. Software Verification:** The 3D reconstructions, point-capture calibrations, and contour-trace workflows executed within the MeViSlab medical image-processing platform were counter-checked for technical accuracy and consistency.

The dataset, pelvimetric values, and morphometric reports compiled from both cohorts are validated as anatomically accurate and radiologically reliable for advanced statistical comparative analysis in fulfilment of the above-mentioned Ph.D. thesis.

Date: _____ **Place:** _____

Name: _____

MD (Radiodiagnosis)

Designation: _____

Department: _____

Institution: _____

Reg. No.: _____

Official Seal / Stamp

Signature & Date